

Explainable Machine Learning for Fintech Adoption and Behavioural Segmentation in Pakistan

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Abstract—Using explainable machine learning (Random Forest and XGBoost with SHAP) and unsupervised segmentation (k-means), this study identifies who is excluded from Pakistan’s digital financial system and, more specifically, where formal access fails to translate into actual digital usage. Using the Global Findex Database 2025 (Pakistan, $n = 1,000$), we construct a four-level Adoption Tier target and test three hypotheses concerning the access-usage gap, the urban-rural adoption divide, and the relative importance of demographic versus access-based predictors. A tuned Random Forest achieves a macro F1 of 0.494 on a held-out test set, and repeated stratified cross-validation (50 folds) confirms this estimate is stable (mean = 0.480, 95% CI: [0.447, 0.521]). An independently-derived k-means cluster comprising 44.3% of the sample shows near-universal mobile phone ownership (98.0%) and majority internet use (62.3%), yet two-thirds remain formally excluded from the financial system – a finding confirmed through two additional, independent diagnostics: a feature-group ablation study, in which removing behavioural variables causes the largest drop in predictive performance while removing access variables slightly *improves* it, and a model error analysis, in which the profile of individuals the classifier mistakenly predicts as digitally engaged closely matches the access-rich, adoption-poor cluster. Bootstrap resampling confirms the three-cluster segmentation is highly stable (Adjusted Rand Index = 0.904, 95% CI: [0.847, 0.982]). Gender emerges as the strongest structural predictor of cluster membership (Cramer’s $V = 0.807$), exceeding the effect size of urban/rural status, diverging from prior Sub-Saharan African findings that access variables dominate adoption. We report calibration analysis showing the model is systematically underconfident for the Excluded class, and a fairness audit showing that naive subgroup accuracy comparisons are confounded by the strong association between gender and outcome in this population – a methodological caveat we argue should generalize to similar financial-inclusion modelling exercises.

Index Terms—fintech adoption, financial inclusion, explainable AI, SHAP, k-means clustering, model calibration, algorithmic fairness, Pakistan, Global Findex

I. INTRODUCTION

Pakistan’s digital financial system has expanded rapidly over the past decade, yet the relationship between formal financial access and actual digital engagement remains poorly understood at the individual level. Existing research on financial inclusion in emerging economies has tended to treat access and usage as functionally equivalent, measuring inclusion primarily through account ownership rates. This conflation obscures a more granular and policy-relevant question: among people who do have access, who actually uses digital financial tools, and who does not?

This study applies explainable machine learning and unsupervised segmentation to individual-level survey data to identify who is excluded from Pakistan’s digital financial system and, more specifically, where formal access fails to translate into actual digital usage. Using Random Forest and XGBoost classifiers with SHAP explainability alongside k-means clustering, the analysis tests three hypotheses concerning the access-usage gap, the urban-rural adoption divide, and the relative importance of demographic versus access-based predictors of adoption. Beyond the core predictive and segmentation results, we subject every major finding to an explicit robustness check: hyperparameter-tuned models are evaluated under repeated cross-validation rather than a single train-test split; the SHAP-based driver ranking is cross-checked against a feature-group ablation study using an entirely different methodology; and the cluster structure is validated via bootstrap resampling and an algorithm-independent comparison against agglomerative clustering. This layered validation approach is, to our knowledge, applied here for the first time to South Asian fintech adoption data, and it surfaces a finding – an access-rich but adoption-poor population segment comprising nearly half the sample – that is corroborated independently by three separate analytical lenses: unsupervised clustering, model error patterns, and ablation-based feature-group importance.

A. Novelty Statement

This work makes four specific contributions beyond prior literature. First, it is the first application of the dual supervised-unsupervised explainable ML methodology of Chowdhury et al. [1] to South Asian survey data, using the newer 2024-fieldwork Findex round rather than the 2021 round used in comparable Sub-Saharan African studies [3]. Second, it identifies and triangulates – via clustering, model error analysis, and ablation – a large access-rich, adoption-poor population segment that is substantially larger than the directly-measured dormant-account rate would suggest, demonstrating the value of combining multiple analytical lenses rather than relying on a single metric. Third, it provides an explicit robustness and calibration audit rarely reported in this literature, showing that headline accuracy figures can mask systematic underconfidence for the most policy-relevant class (Excluded) and that naive subgroup accuracy comparisons can be misleading when a demographic variable is strongly entangled with outcome. Fourth, it offers a direct, quantitative test of whether

the access-dominant driver ranking reported for Sub-Saharan Africa [3] generalizes to Pakistan, finding that it does not – behavioural and demographic variables outrank raw access variables in this context, a result confirmed independently through SHAP, permutation importance, and ablation.

II. RELATED WORK

A. Measuring Financial Inclusion

The Global Findex Database, now in its fifth edition, remains the primary demand-side data source for cross-country financial inclusion research [6]. Early editions of Findex research documented steady global growth in account ownership but repeatedly noted that account ownership alone is an incomplete proxy for financial inclusion, since it does not capture whether an account is actively used. Suri and Jack [5] provided some of the most influential causal evidence on this front, showing via a long-run panel study in Kenya that access to mobile money – specifically, active usage rather than mere availability – reduced extreme poverty and produced disproportionate gains for female-headed households through occupational shifts into business and retail activity. Their finding that gendered impacts of digital financial services can be large and structurally significant directly motivates our attention to gender as a candidate driver of adoption, rather than treating it as a routine demographic control variable.

B. Machine Learning Approaches to Fintech Adoption

Three recent studies directly inform the design of this project. Chowdhury et al. [1] applied Random Forest, XGBoost, and k-means clustering to financial literacy and behaviour data from 1,067 adults in Bangladesh, validating their segmentation with silhouette, Davies-Bouldin, and Calinski-Harabasz indices. They identified three behavioural clusters (34%, 41%, and 25% of the sample) and reported a weak correlation between financial knowledge and financial behaviour, alongside a counterintuitive 7.3 percentage-point rural literacy advantage over urban respondents, with formal education emerging as a comparatively weak predictor relative to institutional trust.

Emuron, Kapim Kenfack, and Msimanga [2] took a more applied approach, predicting fintech adoption in South Africa from Findex, mobile pricing, and central bank data using regression-based methods on remittances, borrowing, age, and digital usage variables. Their work established a directly comparable variable set to the one constructed here from Findex Pakistan, though without a segmentation component or an explicit robustness audit.

Saidy and Hassan [3] modelled fintech adoption drivers across 36 Sub-Saharan African countries using the 2021 Global Findex round, combining Logit and Probit regression with Random Forest, Gradient Boosting, and XGBoost. Their central finding was that formal account ownership and technology access – specifically mobile phone ownership and internet connectivity – were the strongest predictors of adoption, a finding we directly test for generalizability in Section V.

C. Positioning

This project combines Chowdhury et al.’s dual supervised-unsupervised methodology with the adoption-prediction framing of Emuron et al. and Saidy and Hassan, applying both to Pakistan using the newer 2024-fieldwork Findex round [4] – a combination not yet applied to South Asian data. It further extends all three prior works methodologically by adding hyperparameter-tuned robustness estimates, SHAP dependence analysis, calibration diagnostics, cross-method feature importance validation, model error analysis linked back to the segmentation results, an ablation study, and bootstrap-validated cluster stability – none of which are reported in the three anchor papers.

III. DATA AND METHODS

A. Data

The analysis uses the Global Findex Database 2025, Pakistan microdata (Reference: PAK_2024_FINDEX_v02_M) [4], an individual-level survey of 1,000 respondents conducted by the World Bank between May and December 2024. The dataset includes 183 variables spanning demographics, account ownership, digital and informal financial behaviour, and financial resilience. Findex microdata is collected under informed consent protocols administered by the World Bank and its survey partners and is distributed in de-identified form; no personally identifying information is available to or used by this analysis (see Section VII for further discussion).

B. Target Construction

A four-level ordinal Adoption Tier was constructed from account ownership and digital usage indicators: Excluded (no formal account), Basic (account held, no digital usage), Digitally Engaged (uses digital payments and/or a digitally enabled account), and Advanced (digital usage combined with formal saving or borrowing behaviour). The resulting distribution was 68.2% Excluded, 2.1% Basic, 10.1% Digitally Engaged, and 19.6% Advanced, shown in Fig. 1.

C. Feature Selection and Leakage Control

Twenty variables across four construct groups (demographics, access, behaviour, resilience) were selected from the Findex codebook. Because the Adoption Tier target was constructed directly from `account`, `dig_account`, `anydigpayment`, `saved`, and `fin22a` – and because `account_fin` and `account_mob` are near-deterministic components of `account` itself – these seven variables were excluded from the supervised and unsupervised feature sets, leaving fifteen predictors organized into four construct groups (Table I). An initial model run that retained the excluded variables produced spurious near-perfect accuracy, which was identified as data leakage and corrected before further analysis; this correction is reported for methodological transparency.

TABLE I
FEATURE GROUPS USED IN MODELLING AND ABLATION

Group	Variables
Demographics	female, age, educ, inc_q, in_workforce, is_rural
Access	con1, internet_use
Behaviour	merchantpay_dig, fin17b, fin17c, borrowed, fin22a_1
Resilience	fin24a, fin24b

D. Supervised Modelling

Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers were first trained with standard configurations on a stratified 75/25 train-test split, with class weighting applied to address the imbalanced target distribution. The best-performing default configuration (Random Forest) was then re-tuned via `RandomizedSearchCV` (25 iterations, 5-fold stratified cross-validation, scored on macro F1) over a search space covering the number of estimators, maximum depth, minimum samples per leaf, and maximum features considered per split; the full search space is reported in Appendix A. An equivalent search was run for XGBoost for comparison.

E. Model Robustness

To assess whether performance estimates were sensitive to the particular train-test split drawn, the tuned Random Forest was additionally evaluated under repeated stratified 5-fold cross-validation with 10 repeats (50 total folds), yielding a distribution of macro F1 scores rather than a single point estimate.

F. Explainability

The tuned Random Forest was explained using SHAP `TreeExplainer` values. Because the target is multiclass, per-class SHAP arrays were averaged (mean absolute value) to produce an overall importance ranking, and SHAP dependence plots were generated for the top four ranked features against the Advanced-tier class specifically, as this class is the most directly policy-relevant “successful adoption” outcome.

G. Calibration Analysis

Predicted class probabilities were evaluated for calibration using one-vs-rest reliability diagrams (5 quantile-based bins per class) and the Brier score, to assess whether predicted probabilities correspond to observed outcome frequencies – a property distinct from, and not guaranteed by, classification accuracy.

H. Feature Importance Consistency

To assess whether the SHAP-based driver ranking was an artifact of a single explanation method, we additionally computed Gini importance (native to the Random Forest) and permutation importance (20 repeats, scored on macro F1 degradation on the held-out test set), and compared all three rankings via Spearman rank correlation.

I. Error Analysis

Misclassified test-set observations were examined for (i) the dominant confusion pair, (ii) subgroup accuracy by gender and urban/rural status, and (iii) the mean feature profile of misclassified versus correctly classified observations within the dominant confusion pair, to determine whether model errors carried substantive meaning rather than representing unstructured noise.

J. Ablation Study

To independently validate the SHAP-based ranking of construct groups, we conducted a feature-group ablation study, retraining the tuned Random Forest configuration on (i) the full feature set, (ii) each construct group individually held out, and (iii) each construct group used in isolation, measuring macro F1 in each configuration.

K. Unsupervised Segmentation

K-means clustering was applied to the same leakage-safe feature set (standardized via z-score) for $k = 2$ to 5, with k selected by silhouette score and Davies-Bouldin and Calinski-Harabasz indices computed for robustness (Fig. 10). Clusters were profiled on demographic and access characteristics and visualized via a two-component PCA projection (Fig. 11).

L. Cluster Validation

Three checks were used to validate the k-means solution beyond the silhouette score alone. First, bootstrap stability was assessed by refitting k-means on 100 bootstrap resamples and computing the Adjusted Rand Index (ARI) between each resample’s cluster assignment (applied to the full original sample) and the original clustering. Second, an algorithm-independent cross-check compared the k-means solution against Ward agglomerative clustering at the same k . Third, because the silhouette score for $k = 3$ was only marginally higher than for $k = 2$ (0.1675 vs. 0.1674), we conducted an explicit interpretability comparison between the two solutions, examining what population-level distinction the additional cluster does or does not preserve.

M. Statistical Validation

Chi-square tests of independence with Cramer’s V effect sizes were used to test associations between cluster membership, Adoption Tier, gender, and urban/rural status. Wilson score 95% confidence intervals were computed for all headline proportions, given the small size of some subgroups (e.g. $n = 21$ for the Basic tier).

IV. RESULTS

A. H1 – The Access-Usage Gap

Among the 318 respondents holding a formal account, 6.6% (95% CI: 4.4%–9.9%) showed no digital financial usage – a direct measure of the dormant-account phenomenon described by Chowdhury et al. [1] as a knowledge-behaviour gap in the Bangladesh context.

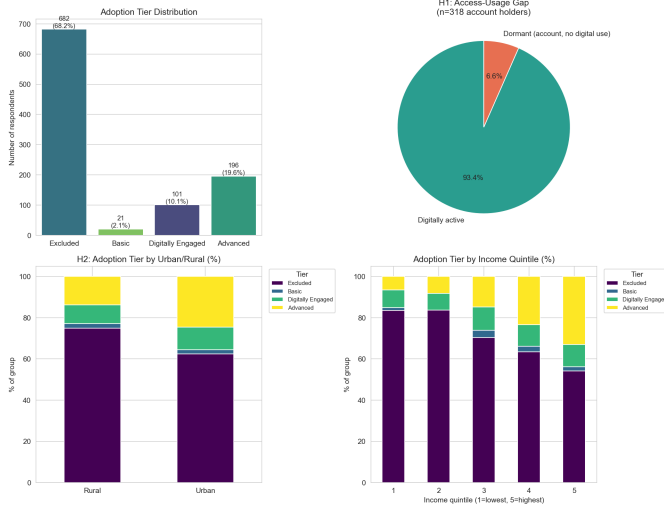


Fig. 1. Adoption Tier distribution, H1 access-usage gap, and Adoption Tier by urban/rural and income quintile.

A substantially larger and more striking pattern emerged from the independent, unsupervised cluster analysis. K-means ($k = 3$) identified a cluster – labelled “Connected but Unengaged” – comprising 44.3% of the entire sample ($n = 443$). This cluster shows near-universal mobile phone ownership (98.0%) and majority internet use (62.3%), yet 66.8% (95% CI: 62.3%–71.0%) remain formally Excluded from the financial system (Table VI). Because this pattern emerged from clustering on access and demographic variables without reference to the Adoption Tier labels, it constitutes independent confirmation that access is not the binding constraint for a large share of the population.

B. H2 – Urban versus Rural Adoption

Adoption Tier was significantly associated with urban/rural status ($\chi^2 = 21.742$, $p = 0.0001$, $df = 3$, Cramer’s $V = 0.147$), and cluster membership showed a parallel association ($\chi^2 = 19.823$, $p < 0.0001$, $df = 2$, Cramer’s $V = 0.141$). The direction of the effect favoured urban respondents: rural share declined monotonically across clusters of increasing adoption, from 51.0% in the least-adopted cluster to 46.0% and then 28.6% in progressively more digitally engaged clusters. This is consistent with the conventional urban-adoption advantage reported broadly in the financial inclusion literature, and contrasts with Chowdhury et al.’s reversed finding of a rural literacy advantage in Bangladesh [1].

C. H3 – Predictors of Adoption

Model comparison results (Table II) show XGBoost achieved the highest single-split accuracy (0.800) but Random Forest achieved the highest macro F1 (0.486 default; 0.494 tuned), the metric prioritized here given substantial class imbalance in the target. All models struggled with the Basic tier ($n = 21$), attributable to sample size rather than model deficiency (Fig. 2).

TABLE II
SUPERVISED MODEL COMPARISON

Model	Accuracy	F1 (macro)	F1 (weighted)
XGBoost	0.800	0.481	0.781
Random Forest (default)	0.708	0.486	0.737
Random Forest (tuned)	–	0.494	–
Decision Tree	0.488	0.424	0.554
Logistic Regression	0.420	0.392	0.473

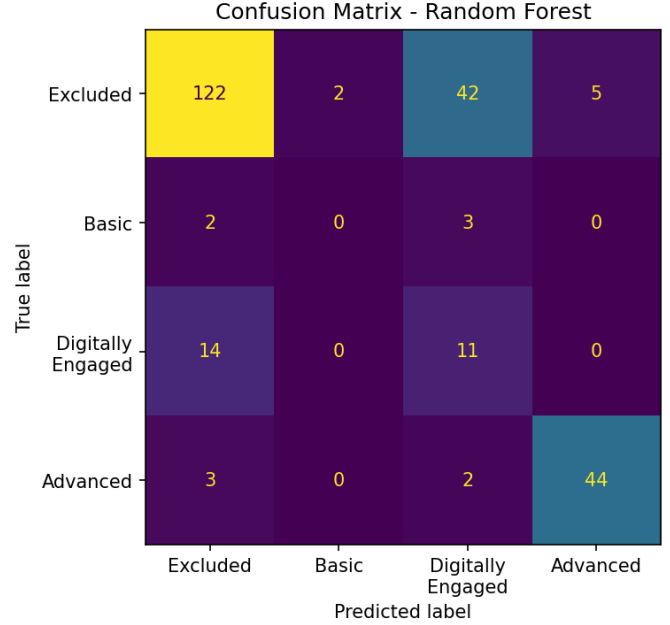


Fig. 2. Confusion matrix for the best-performing model (Random Forest).

SHAP analysis of the tuned Random Forest (Fig. 4) ranked predictors as follows: saving via mobile money ($fin17b$), age, workforce participation, informal saving ($fin17c$), income quintile, mobile phone ownership, emergency fund coverage time, gender, and internet use. Mobile phone ownership and internet use – the two strongest predictors identified by Saidy and Hassan across Sub-Saharan Africa [3] – ranked in the middle of the importance distribution rather than at the top.

1) *SHAP Dependence Plots*: Fig. 5 shows dependence plots for the four top-ranked features against the Advanced-tier class. $fin17b$ and $fin17c$ show clean binary step functions, as expected for binary indicators. Notably, age shows a nonlinear relationship: SHAP contribution rises through the twenties and thirties, peaks around ages 40–50, and declines thereafter – indicating that adoption is not simply monotonic in age, but concentrated among economically active, middle-aged respondents.

2) *Calibration Analysis*: Fig. 6 shows per-class reliability diagrams. The Excluded class is poorly calibrated and systematically underconfident (Brier = 0.168): when the model assigns roughly 40% probability to Excluded, the true observed frequency is closer to 65–70%. The Advanced class

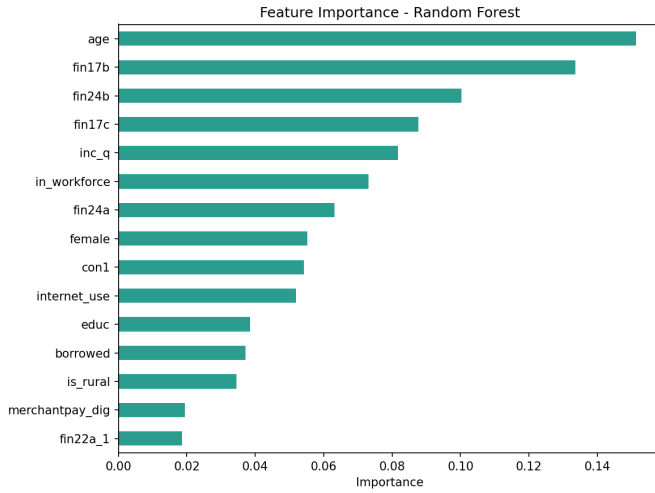


Fig. 3. Random Forest feature importance (Gini-based).

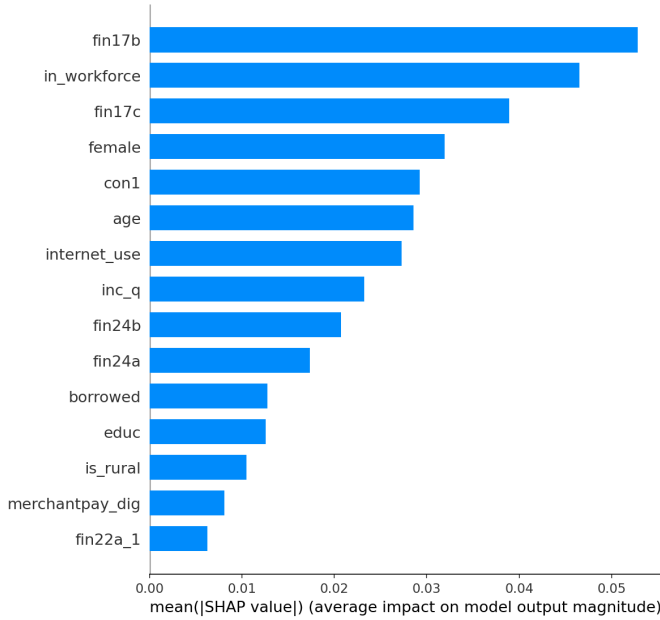


Fig. 4. Mean absolute SHAP value per feature, averaged across all four Adoption Tier classes.

is comparatively well-calibrated (Brier = 0.042), closely tracking the diagonal. The Basic and Digitally Engaged classes show jagged, unreliable calibration curves, consistent with their small effective sample sizes.

3) *Feature Importance Consistency*: Table III and Fig. 7 compare Gini, permutation, and SHAP importance rankings. Gini and SHAP rankings agree strongly (Spearman $\rho = 0.914$, $p < 0.0001$); permutation importance agrees more weakly with SHAP ($\rho = 0.504$, $p = 0.056$, not significant at $\alpha = 0.05$) and is negative for several features (con1, female, internet_use, fin24a, educ, is_rural), indicating that shuffling these features does not reliably degrade – and can marginally improve – macro F1 on the test set. fin17b

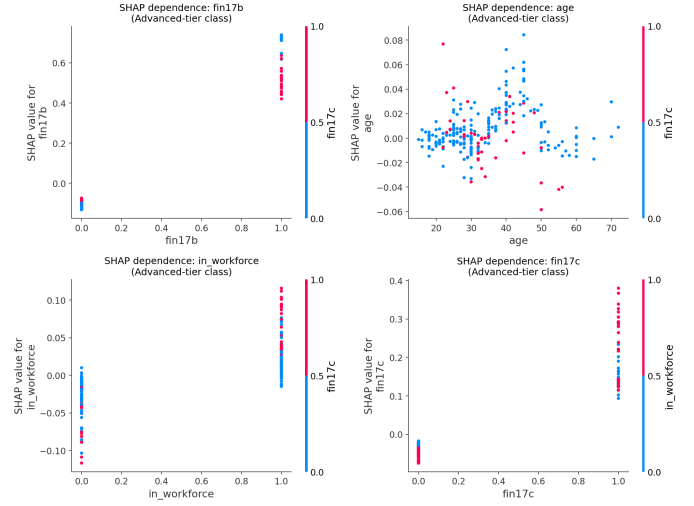


Fig. 5. SHAP dependence plots for the top four features (fin17b, age, in_workforce, fin17c) against the Advanced-tier class, colored by the strongest interacting feature.

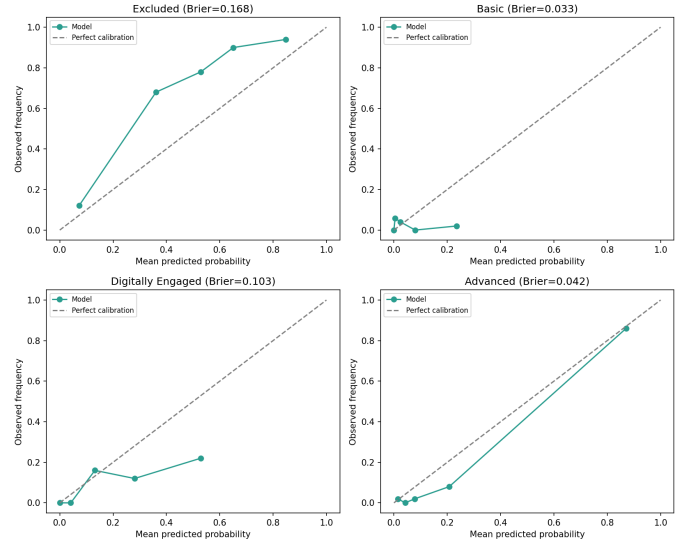


Fig. 6. Per-class reliability diagrams (one-vs-rest) with Brier scores.

is the only feature ranked at or near the top by all three methods, making it the single most robust predictor identified in this analysis, subject to the routing-pattern caveat discussed in Section VI.

4) *Error Analysis*: The dominant confusion pair was Excluded predicted as Digitally Engaged (21 cases), with the reverse error (Digitally Engaged predicted as Excluded, 16 cases) also common – consistent with this boundary being the hardest to separate given the available features.

Subgroup accuracy was 0.852 for women versus 0.689 for men (ratio 0.808), and 0.780 for rural versus 0.752 for urban respondents (ratio 0.964). We caution against interpreting the gender gap as evidence the model treats women “more fairly”: because gender is very strongly associated with cluster membership and outcome in this population (Cramer’s

TABLE III
TOP FEATURES BY CROSS-METHOD IMPORTANCE (ABBREVIATED; FULL
TABLE IN APPENDIX B)

Feature	Gini rank	Permutation rank	SHAP rank
fin17b	2	1	1
age	1	7	2
in_workforce	5	3	3
fin17c	6	2	4
inc_q	3	6	5

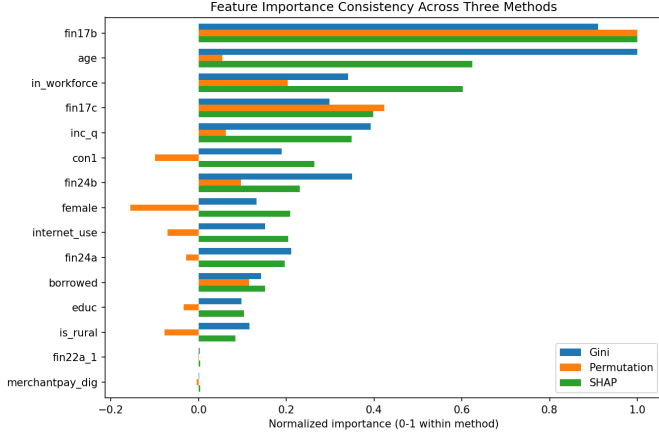


Fig. 7. Normalized feature importance across Gini, permutation, and SHAP methods.

$V = 0.807$; see Section V-F), the majority of women fall into the large, easy-to-predict Excluded class, mechanically inflating aggregate accuracy for that subgroup. We regard this as an important methodological caveat for fairness auditing in financial-inclusion modelling more broadly, discussed further in Section VII.

More substantively, we profiled the feature means of individuals with true label Excluded, comparing those correctly classified against those misclassified as Digitally Engaged (Table IV). Misclassified individuals show markedly higher mobile phone ownership (93.5% vs. 49.3%), higher internet use (41.9% vs. 20.7%), higher workforce participation (80.6% vs. 28.6%), and higher income, and are less likely to be female or rural. This profile closely matches the “Connected but Unengaged” cluster identified independently in Section V-A, providing a second, model-error-based confirmation of the H1 finding through an entirely different analytical route.

5) *Ablation Study*: Table V and Fig. 8 show macro F1 under each feature-group configuration. Removing the behaviour group causes the largest performance drop ($0.494 \rightarrow 0.415$, -16.1%), while removing the access group slightly *improves* macro F1 ($0.494 \rightarrow 0.506$). Access-only as the sole feature group performs worst of all configurations (macro F1 = 0.186, barely above a majority-class baseline). This provides a second, methodologically independent confirmation of the SHAP-based ranking in Section V-C: behavioural variables carry more predictive weight than raw access variables in this population.

TABLE IV
FEATURE MEANS FOR TRUE-EXCLUDED CASES: CORRECT VS.
MISCLASSIFIED

Feature	Misclassified	Correctly Classified
% Female	12.9%	69.3%
Mean age	38.1	33.8
% In workforce	80.6%	28.6%
% Rural	38.7%	51.4%
% Mobile phone (con1)	93.5%	49.3%
% Internet use	41.9%	20.7%
Mean income quintile	3.32	3.13

TABLE V
ABLATION STUDY RESULTS

Configuration	n features	Macro F1
Full feature set	15	0.494
All except demographics	9	0.480
All except access	13	0.506
All except behaviour	10	0.415
All except resilience	13	0.489
Demographics only	6	0.351
Access only	2	0.186
Behaviour only	5	0.389
Resilience only	2	0.305

D. Model Robustness

Repeated stratified cross-validation (50 folds) of the tuned Random Forest yielded a macro F1 distribution with mean = 0.480, std = 0.023, and 95% percentile interval [0.447, 0.521] (Fig. 9), tightly bracketing the single-split test estimate of 0.494 and indicating the reported performance is not an artifact of one particular train-test partition.

E. Cluster Profiles and Gender Association

This gender effect noted in Section V-D-4 is confirmed with substantially larger magnitude at the cluster level. Cluster membership was very strongly associated with gender ($\chi^2 = 650.549$, $p < 0.0001$, $df = 2$, Cramer’s $V = 0.807$), and Adoption Tier itself showed a large association with gender ($\chi^2 = 162.080$, $p < 0.0001$, $df = 3$, Cramer’s $V = 0.403$) – both well above the effect sizes observed for urban/rural status. The most digitally excluded cluster was 96.3% female (95% CI: 94.1%–97.7%), compared to 16.3% and 10.3% female in the two more digitally engaged clusters (Table VI).

F. Cluster Validation

Bootstrap resampling (100 replicates) showed the $k = 3$ solution is highly stable: mean ARI = 0.904 (std = 0.041), 95% percentile interval [0.847, 0.982] (Fig. 13). An algorithm-independent cross-check against Ward agglomerative clustering at the same k showed substantial agreement (ARI = 0.740), indicating the three-cluster structure is not an artifact of the k-means algorithm specifically.

Despite the marginal silhouette difference between $k = 2$ (0.1675) and $k = 3$ (0.1675), the two solutions are substantively different (ARI = 0.623). The $k = 2$ solution separates

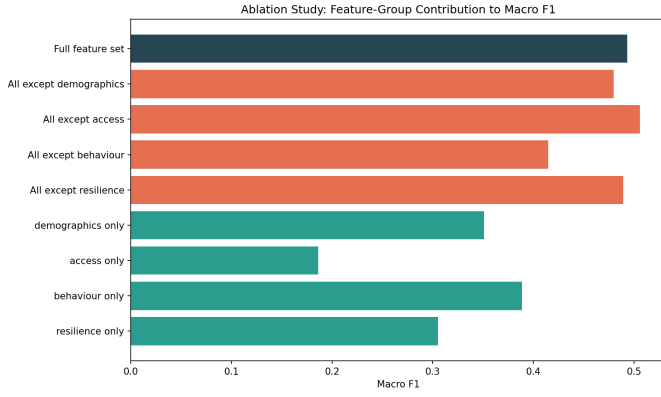


Fig. 8. Ablation study: macro F1 under feature-group leave-one-out and single-group-only configurations.

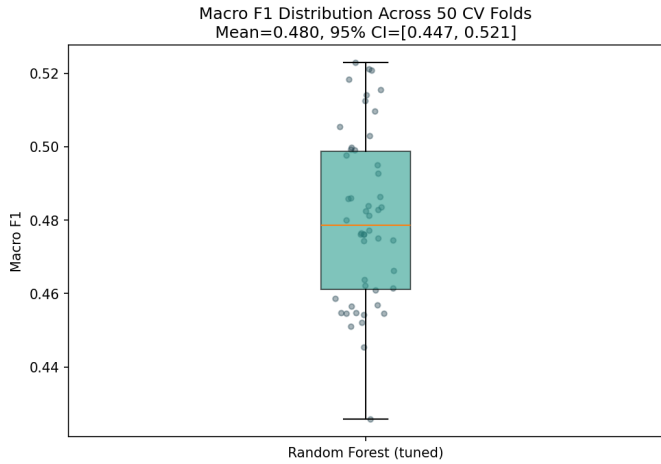


Fig. 9. Distribution of macro F1 across 50 repeated stratified cross-validation folds.

the sample almost entirely along the “Digitally Excluded Women” cluster versus everyone else (515 of 516 members of one $k = 2$ cluster correspond to the original clusters 1 and 2 combined). What $k = 3$ adds is a further split of that “everyone else” group into the access-rich-but-unengaged cluster (66.8% still Excluded) versus the advanced-adopter cluster (0.8% Excluded) – a distinction directly relevant to the access-usage gap central to H1, and the primary substantive justification for reporting $k = 3$ despite its narrow quantitative margin over $k = 2$.

V. DISCUSSION

The central contribution of this analysis is the identification of a large, access-rich but adoption-poor population segment, discovered independently through three separate analytical routes: unsupervised clustering (Section V-A), feature-group ablation (Section V-D-6), and model error analysis (Section V-D-5). At 44.3% of the sample, this segment is far larger than the 6.6% dormant-account rate would suggest, and it reframes the practical question facing Pakistan’s digital finance sector: the primary obstacle to inclusion is not, for a substantial share

TABLE VI
CLUSTER PROFILES (K-MEANS, $K=3$)

Cluster	n (%)	% Female	% Rural	% Mobile	% Excluded
0 – Digitally Excluded Women	431 (43.1%)	96.3%	51.0%	28.3%	89.3%
1 – Connected but Unengaged	443 (44.3%)	16.3%	46.0%	98.0%	66.8%
2 – Advanced Adopters	126 (12.6%)	10.3%	28.6%	98.4%	0.8%

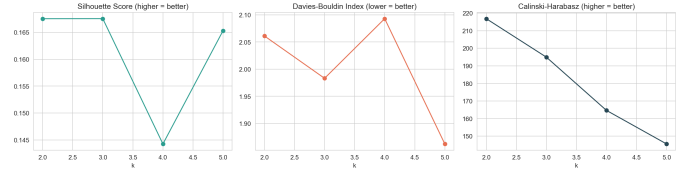


Fig. 10. Silhouette, Davies-Bouldin, and Calinski-Harabasz scores for $k = 2$ to 5.

of the population, a lack of phones or internet connectivity, but something downstream of that access – plausibly trust, product design, documentation requirements, or social/institutional barriers not captured directly in this dataset. The convergence of three independent methods on the same population segment substantially strengthens confidence in this finding relative to any single method in isolation.

The gender findings compound this picture. Rather than a marginal demographic control variable, gender shows one of the largest effect sizes in the entire analysis (Cramer’s $V = 0.807$ for cluster membership), exceeding even the urban/rural divide by a wide margin, and echoes the gendered impacts of mobile money access documented by Suri and Jack [5] in a causal, longitudinal Kenyan setting. Our cross-sectional Pakistani data cannot establish causality in either direction, but the consistency of a large gender effect across two very different national contexts and methodologies is notable. This diverges from Saidy and Hassan’s Sub-Saharan African findings, where account ownership and technology access dominated [3], and suggests that regional context materially changes which factors predict fintech adoption – access-centric interventions calibrated for Sub-Saharan Africa may be poorly suited to Pakistan, where the access-usage gap and the gender gap appear to be entangled with each other rather than separate problems requiring separate interventions.

The H2 finding – urban adoption exceeding rural adoption – aligns with the mainstream financial inclusion literature but diverges from Chowdhury et al.’s Bangladesh result [1]. Since Chowdhury et al. measured financial literacy rather than behavioural adoption, this divergence may reflect a genuine difference between knowledge and behaviour as outcome variables, rather than a contradiction between the two country contexts.

The calibration results carry a practical implication distinct from the accuracy and F1 results typically reported in this literature: a model with reasonable discriminative performance (macro F1 ≈ 0.49) can still be poorly calibrated for its most policy-relevant class. Here, the model systematically underestimates the probability of exclusion, which would



Fig. 11. PCA projection of k-means clusters ($k = 3$).

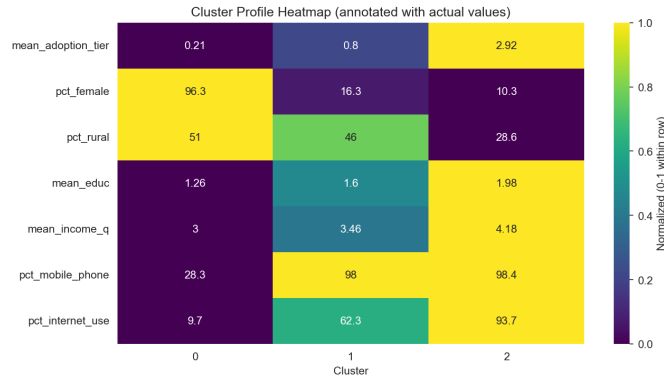


Fig. 12. Normalized cluster profile heatmap across demographic and access variables.

understate the scale of the excluded population if predicted probabilities, rather than hard class labels, were used directly for resource targeting. This is a methodological point we believe deserves more attention in applied fintech-adoption modelling generally, where headline accuracy is often reported without accompanying calibration diagnostics.

The feature importance consistency results similarly complicate a simple “top predictor” narrative. While Gini and SHAP importance agree strongly, permutation importance – arguably the most directly interpretable of the three, since it measures actual performance degradation – agrees only weakly and assigns negative importance to several features including gender and rural status. We interpret this as evidence that Random Forest’s internal (Gini) importance and SHAP’s coalition-based attributions can overstate the standalone predictive necessity of a feature when that feature is correlated with others already carrying the relevant signal (e.g., gender’s correlation with the cluster structure itself), a distinction between *attribution* and *necessity* that we believe is under-discussed in applied SHAP-based fintech literature.

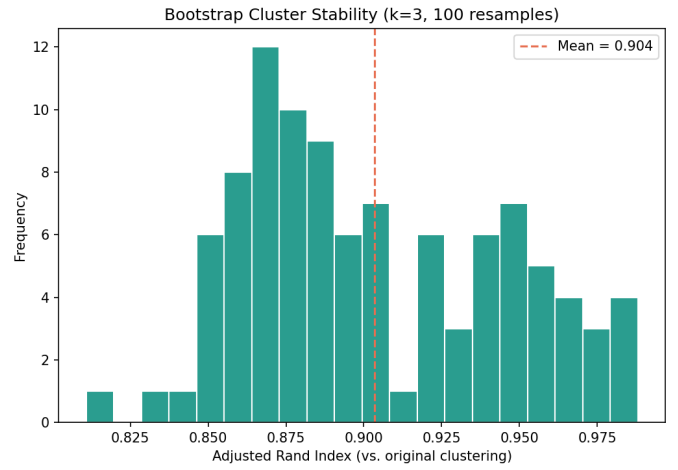


Fig. 13. Distribution of Adjusted Rand Index across 100 bootstrap resamples, evaluating stability of the $k=3$ clustering solution.

VI. ETHICAL CONSIDERATIONS

A. Fairness and the Accuracy Paradox

Section V-D-5 demonstrated that raw subgroup accuracy (0.852 for women vs. 0.689 for men) is a misleading fairness metric in this setting, because gender is strongly entangled with the outcome variable itself at the population level. A naïve fairness audit relying on aggregate accuracy would conclude the model treats women more favorably, when in fact women are disproportionately concentrated in the largest, easiest-to-predict class. We recommend that fairness audits of financial-inclusion models report subgroup-level macro F1 or per-class recall rather than aggregate accuracy, and we flag this as a specific methodological risk for future work in this domain, since account ownership and gender are correlated by construction in many emerging-market contexts, not just Pakistan.

B. Risk of Misuse

Adoption-tier prediction models of the type developed here are intended to support inclusion-oriented policy and product design – for example, identifying the access-rich, adoption-poor segment as a target for trust-building or product-design interventions. The same predictive infrastructure could, however, be repurposed for exclusionary rather than inclusionary ends: for instance, using predicted low adoption likelihood as an input to credit-scoring or loan-eligibility decisions, which would risk encoding and amplifying existing gender and rural disparities into automated lending decisions. We explicitly do not recommend this model, or models like it, be used for individual-level credit or eligibility decisions; its intended use is aggregate, population-level policy targeting.

C. Calibration and Deployment Caution

Given the poor calibration observed for the Excluded class (Section V-D-3), we caution against using this model’s predicted probabilities directly in any downstream resource-allocation formula without recalibration (e.g., Platt scaling

or isotonic regression), since the systematic underconfidence identified here would bias such a formula toward underestimating exclusion.

D. Data Privacy

All analysis was conducted on de-identified, publicly available World Bank microdata collected under the Global Findex survey’s standard informed-consent protocols. No attempt was made, or would be possible, to re-identify individual respondents from the released microdata.

VII. LIMITATIONS

- The Basic tier ($n = 21$) is too small for reliable classification by any model tested; this is a sample-size constraint rather than a modelling failure and should not be over-interpreted.
- The variable `fin17b` (saved via mobile money), the top-ranked predictor by both Gini and SHAP, is partly confounded with mobile money account ownership due to Findex’s survey routing logic (approximately 74% of respondents were routed out of this question); its top ranking should be interpreted with this caveat.
- Permutation importance disagreed substantively with Gini and SHAP for several features, including assigning negative importance to gender and rural status; we interpret this as reflecting correlated-feature attribution effects rather than concluding these features are unimportant, but this disagreement itself is a limitation of relying on any single importance method.
- The silhouette score for $k = 3$ was only marginally higher than for $k = 2$ (Section V-F), indicating the three-cluster solution, while interpretable and independently validated via bootstrap and algorithm cross-checks, is not an overwhelmingly dominant statistical structure by silhouette score alone.
- Calibration is poor for the Excluded, Basic, and Digitally Engaged classes; only the Advanced class is well-calibrated. Predicted probabilities for the other three classes should not be used without recalibration.
- Several chi-square tests involve subgroups with small expected cell counts (notably comparisons involving the Basic tier); results for these specific comparisons should be treated as indicative rather than definitive.
- Subgroup accuracy is confounded by the strong association between gender and outcome in this sample, as discussed in Section VII-A; this is a general risk for fairness auditing in similar settings, not specific to a modelling error here.
- All findings are associational. Findex is a cross-sectional survey, and no causal claims are made regarding the drivers of adoption or the mechanisms underlying the gender and access-usage gaps identified.

VIII. FUTURE WORK

Several extensions would strengthen this analysis. First, replicating the pipeline on the Financial Inclusion Insights

(FII) Bangladesh tracker survey, if obtained, would provide a genuine same-instrument cross-country comparison. Second, a qualitative or mixed-methods follow-up focused specifically on the “Connected but Unengaged” cluster could clarify which downstream barriers – trust, documentation, product design – best explain non-adoption despite access. Third, model recalibration (Platt scaling or isotonic regression) applied specifically to the Excluded class would address the systematic underconfidence identified in Section V-D-3 and should be evaluated before any applied use of predicted probabilities. Fourth, a fairness-aware modelling approach (e.g., equalized-odds post-processing) could be evaluated as a remedy for the subgroup accuracy confound identified in Section VII-A, alongside subgroup-disaggregated macro F1 reporting as a standard diagnostic. Fifth, causal identification strategies (e.g., difference-in-differences around a policy change, or an instrumental-variables approach) would be needed to move beyond the associational claims made throughout this study.

IX. CONCLUSION

This study applied explainable machine learning and unsupervised segmentation, together with an extensive robustness and validation suite, to individual-level Findex data on financial inclusion in Pakistan. The central finding – that 44.3% of the sampled population has near-universal mobile phone and majority internet access yet remains predominantly formally excluded from the financial system – was independently corroborated through clustering, ablation, and model error analysis, and the cluster structure itself was validated as stable under bootstrap resampling and robust to the choice of clustering algorithm. Gender emerged as a structurally significant predictor of both adoption and cluster membership, with an effect size exceeding that of urban/rural status, diverging from prior Sub-Saharan African evidence that access variables dominate adoption. Calibration and fairness diagnostics further revealed that headline performance metrics can obscure systematic underconfidence for policy-relevant classes and can produce misleading subgroup fairness comparisons when a demographic variable is entangled with outcome – methodological cautions we believe merit wider adoption in applied fintech-adoption modelling.

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APPENDIX A HYPERPARAMETER SEARCH SPACE

TABLE VII
HYPERPARAMETER SEARCH SPACE AND SELECTED VALUES

Model	Parameter	Search space (selected)
Random Forest	n_estimators	{200, 300, 500, 800} (300)
	max_depth	{4, 6, 8, 10, None} (None)
	min_samples_leaf	{1, 2, 4, 8} (4)
	max_features	{sqrt, log2, None} (None)
XGBoost	n_estimators	{200, 300, 500} (500)
	max_depth	{3, 4, 5, 6} (6)
	learning_rate	{0.01, 0.03, 0.05, 0.1} (0.1)
	subsample	{0.7, 0.85, 1.0} (0.85)
	colsample_bytree	{0.7, 0.85, 1.0} (0.85)

Table VII lists the search space used for RandomizedSearchCV (25 iterations, 5-fold stratified CV, scored on macro F1) and the selected best configuration for each model.

Best cross-validated macro F1: Random Forest = 0.472; XGBoost = 0.457. Held-out test macro F1: default Random Forest = 0.486; tuned Random Forest = 0.494.

APPENDIX B ADDITIONAL DIAGNOSTIC TABLES

Table VIII reports the complete feature importance comparison across all fifteen features and three methods (Gini, permutation, SHAP), supplementing the abbreviated version in Table III.

Spearman rank correlations: Gini vs. SHAP $\rho = 0.914$ ($p < 0.0001$); Permutation vs. SHAP $\rho = 0.504$ ($p = 0.0557$); Gini vs. Permutation $\rho = 0.554$ ($p = 0.0323$).

Table IX reports the complete confusion matrix for the tuned Random Forest on the held-out test set ($n = 250$).

Table X summarizes the cluster validation metrics reported in Section V-G.

TABLE VIII
FULL FEATURE IMPORTANCE COMPARISON

Feature	Gini	Permutation	SHAP
fin17b	0.2099	0.1087	0.0915
age	0.2307	0.0059	0.0571
in_workforce	0.0786	0.0221	0.0551
fin17c	0.0689	0.0459	0.0364
inc_q	0.0905	0.0067	0.0319
con1	0.0437	−0.0108	0.0242
fin24b	0.0808	0.0105	0.0211
female	0.0306	−0.0169	0.0191
internet_use	0.0349	−0.0077	0.0187
fin24a	0.0486	−0.0031	0.0180
borrowed	0.0329	0.0125	0.0139
educ	0.0225	−0.0038	0.0094
is_rural	0.0266	−0.0085	0.0077
fin22a_1	0.0005	−0.0001	0.0003
merchantpay_dig	0.0004	−0.0005	0.0003

TABLE IX
FULL CONFUSION MATRIX (TUNED RANDOM FOREST, TEST SET)

True \ Predicted	Excluded	Basic	Dig. Engaged	Advanced
Excluded	140	3	21	7
Basic	3	0	2	0
Digitally Engaged	16	0	8	1
Advanced	4	0	2	43

TABLE X
CLUSTER VALIDATION SUMMARY

Metric	Value
Bootstrap ARI, mean (std)	0.904 (0.041)
Bootstrap ARI, 95% interval	[0.847, 0.982]
ARI vs. Ward agglomerative	0.740
ARI, $k=2$ vs. $k=3$ solution	0.623
Silhouette score, $k=2$	0.1675
Silhouette score, $k=3$	0.1675